

DER Models

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1 Without Storage

1.1 Disaggregation

$$\max \sum_{t \in T} (P_t^{DA} x_{it} + \mathbb{E} [P_t^{RT}(\xi) y_{it}^+(\xi) - P_t^{PN} y_{it}^-(\xi)]) \quad (1a)$$

$$\text{s.t. } R_{it}(\xi) - x_{it} = y_{it}^+(\xi) - y_{it}^-(\xi) \quad \forall t \in T \quad (1b)$$

$$R_{it}(\xi) \geq y_{it}^+(\xi) \quad \forall t \in T \quad (1c)$$

$$y_{it}^+(\xi) \leq M z_{it}(\xi), \quad y_{it}^-(\xi) \leq M(1 - z_{it}(\xi)) \quad \forall t \in T \quad (1d)$$

$$x_{it}^{DA} \geq 0, y_{it}^+(\xi) \geq 0, y_{it}^-(\xi) \geq 0, z_{it}(\xi) \in \{0, 1\} \quad \forall t \in T \quad (1e)$$

1.2 Aggregation

$$\max \sum_{t \in T} (P_t^{DA} \alpha_t + \mathbb{E} [P_t^{RT}(\xi) \beta_t^+(\xi) - P_t^{PN} \beta_t^-(\xi)]) \quad (2a)$$

$$\text{s.t. } \sum_{i \in I} R_{it}(\xi) - \alpha_t = \beta_t^+(\xi) - \beta_t^-(\xi) \quad \forall t \in T \quad (2b)$$

$$\sum_{i \in I} R_{it}(\xi) \geq \beta_t^+(\xi) \quad \forall t \in T \quad (2c)$$

$$\beta_t^+(\xi) \leq M z_t(\xi), \quad \beta_t^-(\xi) \leq M(1 - z_t(\xi)) \quad \forall t \in T \quad (2d)$$

$$\alpha_t^{DA} \geq 0, \beta_t^+(\xi) \geq 0, \beta_t^-(\xi) \geq 0, z_t(\xi) \in \{0, 1\} \quad \forall t \in T \quad (2e)$$

1.3 Settlement

$$\max \sum_{t \in T} (P_t^{DA} \alpha_t + \mathbb{E} [P_t^{RT}(\xi) \beta_t^+(\xi) - P_t^{PN} \beta_t^-(\xi)]) \quad (3a)$$

$$\text{s.t. } \sum_{i \in I} R_{it}(\xi) - \alpha_t^{DA} = \beta_t^+(\xi) - \beta_t^-(\xi) \quad \forall t \in T \quad (3b)$$

$$\sum_{i \in I} R_{it}(\xi) \geq \beta_t^+(\xi) \quad \forall t \in T \quad (3c)$$

$$\beta_t^+(\xi) \leq M z_t(\xi), \quad \beta_t^-(\xi) \leq M(1 - z_t(\xi)) \quad \forall t \in T \quad (3d)$$

$$\alpha_t = \sum_{i \in I} x_{it}(\xi), \quad \beta_t^+(\xi) = \sum_{i \in I} e_{it}^+(\xi), \quad \beta_t^-(\xi) = \sum_{i \in I} e_{it}^-(\xi) \quad \forall t \in T \quad (3e)$$

$$R_{it}(\xi) - x_{it}(\xi) = y_{it}^+(\xi) - y_{it}^-(\xi) \quad \forall t \in T \quad (3f)$$

$$R_{it}(\xi) \geq y_{it}^+(\xi) \quad \forall t \in T \quad (3g)$$

$$y_{it}^+(\xi) \leq M z_{it}(\xi), \quad y_{it}^-(\xi) \leq M(1 - z_{it}(\xi)) \quad \forall t \in T \quad (3h)$$

$$\sum_{j \in I, j \neq i} d_{ijt}(\xi) \leq y_{it}^+(\xi), \quad \sum_{j \in I, j \neq i} d_{jit}(\xi) \leq y_{it}^-(\xi) \quad \forall t \in T \quad (3i)$$

$$d_{iit}(\xi) = 0 \quad \forall t \in T \quad (3j)$$

$$e_{it}^+(\xi) = y_{it}^+(\xi) - \sum_{j \in I, j \neq i} d_{ijt}(\xi) \quad \forall t \in T \quad (3k)$$

$$e_{it}^-(\xi) = y_{it}^-(\xi) - \sum_{j \in I, j \neq i} d_{jit}(\xi) \quad \forall t \in T \quad (3l)$$

2 With Storage

2.1 Disaggregation

$$\max \sum_{t \in T} (P_t^{DA} x_{it} + \mathbb{E} [P_t^{RT}(\xi) y_{it}^+(\xi) - P_t^{PN} y_{it}^-(\xi)]) \quad (4a)$$

$$\text{s.t. } R_{it}(\xi) - x_{it} = y_{it}^+(\xi) - y_{it}^-(\xi) + z_{it}^C(\xi) - z_{it}^D(\xi) \quad (4b)$$

$$R_{it}(\xi) \geq y_{it}^+(\xi) \quad (4c)$$

$$z_{i,t+1}(\xi) = z_{it}(\xi) + z_{it}^C(\xi) - z_{it}^D(\xi) \quad \forall i, t \quad (4d)$$

$$z_{it}^D(\xi) \leq z_{it}(\xi), \quad z_{it}^C(\xi) \leq K_i - z_{it}(\xi), \quad 0 \leq z_{it}(\xi) \leq K_i \quad (4e)$$

$$y_{it}^+(\xi) \leq M_1 \phi_{it}^1(\xi), \quad y_{it}^-(\xi) \leq M_1 (1 - \phi_{it}^1(\xi)) \quad (4f)$$

$$y_{it}^-(\xi) \leq M_1 \phi_{it}^2(\xi), \quad z_{it}^C(\xi) \leq M_1 (1 - \phi_{it}^2(\xi)) \quad (4g)$$

$$z_{it}^C(\xi) \leq M_1 \phi_{it}^3(\xi), \quad z_{it}^D(\xi) \leq M_1 (1 - \phi_{it}^3(\xi)) \quad (4h)$$

2.2 Aggregation with individual BTM storage control - y, d, e correlated

$$\max \sum_{t \in T} \left(P_t^{DA} \sum_{i \in I} x_{it} + \mathbb{E} \left[P_t^{RT}(\xi) \sum_{i \in I} e_{it}^+(\xi) - P_t^{PN} \sum_{i \in I} e_{it}^-(\xi) \right] \right) \quad (5a)$$

$$\text{s.t. } R_{it}(\xi) - x_{it} = y_{it}^+(\xi) - y_{it}^-(\xi) + z_{it}^C(\xi) - z_{it}^D(\xi) \quad (5b)$$

$$R_{it}(\xi) + z_{it}^D(\xi) \geq y_{it}^+(\xi) + z_{it}^C(\xi) \quad (5c)$$

$$z_{i,t+1}(\xi) = z_{it}(\xi) + \eta^C z_{it}^C(\xi) - \frac{1}{\eta^D} z_{it}^D(\xi) \quad (5d)$$

$$z_{it}^D(\xi) \leq z_{it}(\xi), \quad z_{it}^C(\xi) \leq K_i - z_{it}(\xi), \quad 0 \leq z_{it}(\xi) \leq K_i \quad (5e)$$

$$e_{it}^+(\xi) = y_{it}^+(\xi) - d_{it}^+(\xi), \quad e_{it}^-(\xi) = y_{it}^-(\xi) - d_{it}^-(\xi) \quad (5f)$$

$$\sum_{i \in I} d_{it}^+(\xi) = \sum_{i \in I} d_{it}^-(\xi) \quad (5g)$$

$$y_{it}^+(\xi) \leq M_1 \phi_{it}^1(\xi), \quad y_{it}^-(\xi) \leq M_1 (1 - \phi_{it}^1(\xi)) \quad (5h)$$

$$y_{it}^-(\xi) \leq M_1 \phi_{it}^2(\xi), \quad z_{it}^C(\xi) \leq M_1 (1 - \phi_{it}^2(\xi)) \quad (5i)$$

$$z_{it}^C(\xi) \leq M_1 \phi_{it}^3(\xi), \quad z_{it}^D(\xi) \leq M_1 (1 - \phi_{it}^3(\xi)) \quad (5j)$$

2.3 Holistic Aggregation with individual BTM storage control - y, d independent

$$\max \sum_{i \in I} \sum_{t \in T} (P_t^{DA} \cdot x_{it} + \mathbb{E} [P_t^{RT}(\xi) \cdot y_{it}^+(\xi) - P_t^{PN}(\xi) \cdot y_{it}^-(\xi)]) \quad (6a)$$

$$\text{s.t. } R_{it}(\xi) - x_{it} = y_{it}^+(\xi) - y_{it}^-(\xi) + d_{it}^+(\xi) - d_{it}^-(\xi) + z_{it}^C(\xi) - z_{it}^D(\xi) \quad \forall i \in I, t \in T \quad (6b)$$

$$R_{it}(\xi) + z_{it}^D(\xi) \geq y_{it}^+(\xi) + d_{it}^+(\xi) + z_{it}^C(\xi) \quad \forall i \in I, t \in T \quad (6c)$$

$$z_{i,t+1}(\xi) = z_{it}(\xi) + \eta^C z_{it}^C(\xi) - \frac{1}{\eta^D} z_{it}^D(\xi) \quad \forall i \in I, t \in T \quad (6d)$$

$$z_{it}^D(\xi) \leq z_{it}(\xi), \quad z_{it}^C(\xi) \leq K_i - z_{it}(\xi), \quad 0 \leq z_{it}(\xi) \leq K_i \quad \forall i \in I, t \in T \quad (6e)$$

$$y_{it}^+(\xi) \leq M \phi_{it}^1(\xi), \quad y_{it}^-(\xi) \leq M (1 - \phi_{it}^1(\xi)) \quad \forall i \in I, t \in T \quad (6f)$$

$$y_{it}^-(\xi) \leq M \phi_{it}^2(\xi), \quad z_{it}^C(\xi) \leq M (1 - \phi_{it}^2(\xi)) \quad \forall i \in I, t \in T \quad (6g)$$

$$z_{it}^C(\xi) \leq M \phi_{it}^3(\xi), \quad z_{it}^D(\xi) \leq M (1 - \phi_{it}^3(\xi)) \quad \forall i \in I, t \in T \quad (6h)$$

$$d_{it}^+(\xi) \leq M \phi_{it}^4(\xi), \quad d_{it}^-(\xi) \leq M (1 - \phi_{it}^4(\xi)) \quad \forall i \in I, t \in T \quad (6i)$$

$$z_{it}^C(\xi) \leq M \phi_{it}^5(\xi), \quad d_{it}^-(\xi) \leq M (1 - \phi_{it}^5(\xi)) \quad \forall i \in I, t \in T \quad (6j)$$

$$y_{it}^+(\xi) \leq M \phi_{it}^6(\xi), \quad d_{it}^-(\xi) \leq M (1 - \phi_{it}^6(\xi)) \quad \forall i \in I, t \in T \quad (6k)$$

$$y_{it}^-(\xi) \leq M \phi_{it}^7(\xi), \quad d_{it}^+(\xi) \leq M (1 - \phi_{it}^7(\xi)) \quad \forall i \in I, t \in T \quad (6l)$$

$$\sum_{i \in I} d_{it}^+(\xi) = \sum_{i \in I} d_{it}^-(\xi) \quad : \quad \lambda_t(\xi) \quad \forall t \in T \quad (6m)$$

$$\Pi_i^{IND} \leq \Pi_i^{HOL} \quad \forall i \in I \quad (6n)$$

2.4 Aggregation with direct control over storage

$$\max \sum_{t \in T} (P_t^{DA} \alpha_t + \mathbb{E} [P_t^{RT}(\xi) \beta_t^+(\xi) - P_t^{PN} \beta_t^-(\xi)]) \quad (7a)$$

$$\text{s.t.} \quad \sum_{i \in I} R_{it}(\xi) - \alpha_t = \beta_t^+(\xi) - \beta_t^-(\xi) + \gamma_t^C(\xi) - \gamma_t^D(\xi) \quad \forall t \quad (7b)$$

$$\sum_{i \in I} R_{it}(\xi) \geq \beta_t^+(\xi) \quad (7c)$$

$$\gamma_t^D(\xi) \leq \gamma_t(\xi), \quad \gamma_t^C(\xi) \leq \sum_{i \in I} K_i - \gamma_t(\xi), \quad 0 \leq \gamma_t(\xi) \leq \sum_{i \in I} K_i \quad \forall t \quad (7d)$$

$$\gamma_{t+1}(\xi) = \gamma_t(\xi) + \gamma_t^C(\xi) - \gamma_t^D(\xi) \quad \forall t \quad (7e)$$

$$\beta_t^+(\xi) \leq M_2 \mu_t(\xi), \quad \beta_t^-(\xi) \leq M_2(1 - \mu_t(\xi)) \quad \forall t \quad (7f)$$

$$\beta_t^-(\xi) \leq M_2 \eta_t(\xi), \quad \gamma_t^C(\xi) \leq M_2(1 - \eta_t(\xi)) \quad \forall t \quad (7g)$$

$$\gamma_t^C(\xi) \leq M_2 \lambda_t(\xi), \quad \gamma_t^D(\xi) \leq M_2(1 - \lambda_t(\xi)) \quad \forall t \quad (7h)$$

3 Individual

3.1 Different Internal Price (Non-linear)

$$\max \sum_{t \in T} (P_t^{DA} \cdot x_t + \mathbb{E} [P_t^{RT}(\xi) \cdot y_t^+(\xi) - P_t^{PN} \cdot y_t^-(\xi) + \rho_t^+(d) \cdot d_t^+(\xi) - \rho_t^-(d) \cdot d_t^-(\xi)]) \quad (8a)$$

$$\text{s.t. } R_t(\xi) - x_t = y_t^+(\xi) - y_t^-(\xi) + d_t^+(\xi) - d_t^-(\xi) + z_t^C(\xi) - z_t^D(\xi) \quad (8b)$$

$$R_t(\xi) + z_t^D(\xi) \geq y_t^+(\xi) + d_t^+(\xi) + z_t^C(\xi) \quad (8c)$$

$$z_{t+1}(\xi) = z_t(\xi) + z_t^C(\xi) - z_t^D(\xi) \quad (8d)$$

$$z_t^D(\xi) \leq z_t(\xi), \quad z_t^C(\xi) \leq K - z_t(\xi), \quad 0 \leq z_t(\xi) \leq K \quad (8e)$$

$$y_t^+(\xi) \leq M\phi_t^1(\xi), \quad y_t^-(\xi) \leq M(1 - \phi_t^1(\xi)) \quad (8f)$$

$$y_t^-(\xi) \leq M\phi_t^2(\xi), \quad z_t^C(\xi) \leq M(1 - \phi_t^2(\xi)) \quad (8g)$$

$$z_t^C(\xi) \leq M\phi_t^3(\xi), \quad z_t^D(\xi) \leq M(1 - \phi_t^3(\xi)) \quad (8h)$$

$$d_t^+(\xi) \leq M\phi_t^4(\xi), \quad d_t^-(\xi) \leq M(1 - \phi_t^4(\xi)) \quad (8i)$$

3.2 Different Internal Price (Stepwise-Linear)

$$\begin{aligned} \max \quad & \sum_{t \in T} \left(P_t^{DA} \cdot x_t + \mathbb{E} \left[P_t^{RT}(\xi) \cdot y_t^+(\xi) - P_t^{PN} \cdot y_t^-(\xi) \right. \right. \\ & \left. \left. + \sum_{b^+ \in B_t^+} \rho_{t,b^+}^+ \cdot (w_{t,b^+}^+(\xi) + u_{t,b^+}^+(\xi) \cdot D_{t,b^+}^{\min+}) - \sum_{b^- \in B_t^-} \rho_{t,b^-}^- \cdot (w_{t,b^-}^-(\xi) + u_{t,b^-}^-(\xi) \cdot D_{t,b^-}^{\min-}) \right] \right) \end{aligned} \quad (9a)$$

$$\text{s.t. } d_t^+(\xi) = \sum_{b^+ \in B_t^+} (w_{t,b^+}^+(\xi) + u_{t,b^+}^+(\xi) \cdot D_{t,b^+}^{\min+}) \quad \forall t \quad (9b)$$

$$0 \leq w_{t,b^+}^+(\xi) \leq u_{t,b^+}^+(\xi) \cdot W_{t,b^+}^{\max+} \quad \forall t, b^+ \quad (9c)$$

$$\sum_{b^+ \in B_t^+} u_{t,b^+}^+(\xi) \leq 1 \quad \forall t \quad (9d)$$

$$d_t^-(\xi) = \sum_{b^- \in B_t^-} (w_{t,b^-}^-(\xi) + u_{t,b^-}^-(\xi) \cdot D_{t,b^-}^{\min-}) \quad \forall t \quad (9e)$$

$$0 \leq w_{t,b^-}^-(\xi) \leq u_{t,b^-}^-(\xi) \cdot W_{t,b^-}^{\max-} \quad \forall t, b^- \quad (9f)$$

$$\sum_{b^- \in B_t^-} u_{t,b^-}^-(\xi) \leq 1 \quad \forall t \quad (9g)$$

$$R_t(\xi) - x_t = y_t^+(\xi) - y_t^-(\xi) + d_t^+(\xi) - d_t^-(\xi) + z_t^C(\xi) - z_t^D(\xi) \quad \forall t \quad (9h)$$

$$R_t(\xi) + z_t^D(\xi) \geq y_t^+(\xi) + d_t^+(\xi) + z_t^C(\xi) \quad \forall t \quad (9i)$$

$$z_{t+1}(\xi) = z_t(\xi) + z_t^C(\xi) - z_t^D(\xi) \quad \forall t \quad (9j)$$

$$z_t^D(\xi) \leq z_t(\xi), \quad z_t^C(\xi) \leq K - z_t(\xi), \quad 0 \leq z_t(\xi) \leq K \quad \forall t \quad (9k)$$

$$y_t^+(\xi) \leq M\phi_t^1(\xi), \quad y_t^-(\xi) \leq M(1 - \phi_t^1(\xi)) \quad \forall t \quad (9l)$$

$$y_t^-(\xi) \leq M\phi_t^2(\xi), \quad z_t^C(\xi) \leq M(1 - \phi_t^2(\xi)) \quad \forall t \quad (9m)$$

$$z_t^C(\xi) \leq M\phi_t^3(\xi), \quad z_t^D(\xi) \leq M(1 - \phi_t^3(\xi)) \quad \forall t \quad (9n)$$

$$d_t^+(\xi) \leq M\phi_t^4(\xi), \quad d_t^-(\xi) \leq M(1 - \phi_t^4(\xi)) \quad \forall t \quad (9o)$$

4 Endogeneous Pricing

4.1 Price as DV

$$\begin{aligned}
\max \quad & \sum_{i \in I} \sum_{t \in T} (P_t^{DA} \cdot x_{it} + \mathbb{E} [P_t^{RT}(\xi) \cdot y_{it}^+(\xi) - P_t^{PN}(\xi) \cdot y_{it}^-(\xi) + \rho_t^+ \cdot d_{it}^+(\xi) - \rho_t^- \cdot d_{it}^-(\xi)]) \quad (10a) \\
\text{s.t.} \quad & R_{it}(\xi) - x_{it} = y_{it}^+(\xi) - y_{it}^-(\xi) + d_{it}^+(\xi) - d_{it}^-(\xi) + z_{it}^C(\xi) - z_{it}^D(\xi) \quad \forall i \in I, t \in T \quad (10b) \\
& R_{it}(\xi) + z_{it}^D(\xi) \geq y_{it}^+(\xi) + d_{it}^+(\xi) + z_{it}^C(\xi) \quad \forall i \in I, t \in T \quad (10c) \\
& z_{i,t+1}(\xi) = z_{it}(\xi) + z_{it}^C(\xi) - z_{it}^D(\xi) \quad \forall i \in I, t \in T \quad (10d) \\
& z_{it}^D(\xi) \leq z_{it}(\xi), \quad z_{it}^C(\xi) \leq K_i - z_{it}(\xi), \quad 0 \leq z_{it}(\xi) \leq K_i \quad \forall i \in I, t \in T \quad (10e) \\
& y_{it}^+(\xi) \leq M\phi_{it}^1(\xi), \quad y_{it}^-(\xi) \leq M(1 - \phi_{it}^1(\xi)) \quad \forall i \in I, t \in T \quad (10f) \\
& y_{it}^-(\xi) \leq M\phi_{it}^2(\xi), \quad z_{it}^C(\xi) \leq M(1 - \phi_{it}^2(\xi)) \quad \forall i \in I, t \in T \quad (10g) \\
& z_{it}^C(\xi) \leq M\phi_{it}^3(\xi), \quad z_{it}^D(\xi) \leq M(1 - \phi_{it}^3(\xi)) \quad \forall i \in I, t \in T \quad (10h) \\
& d_{it}^+(\xi) \leq M\phi_{it}^4(\xi), \quad d_{it}^-(\xi) \leq M(1 - \phi_{it}^4(\xi)) \quad \forall i \in I, t \in T \quad (10i) \\
& z_{it}^C(\xi) \leq M\phi_{it}^5(\xi), \quad d_{it}^-(\xi) \leq M(1 - \phi_{it}^5(\xi)) \quad \forall i \in I, t \in T \quad (10j) \\
& y_{it}^+(\xi) \leq M\phi_{it}^6(\xi), \quad d_{it}^-(\xi) \leq M(1 - \phi_{it}^6(\xi)) \quad \forall i \in I, t \in T \quad (10k) \\
& y_{it}^-(\xi) \leq M\phi_{it}^7(\xi), \quad d_{it}^+(\xi) \leq M(1 - \phi_{it}^7(\xi)) \quad \forall i \in I, t \in T \quad (10l) \\
& \sum_{i \in I} d_{it}^+(\xi) = \sum_{i \in I} d_{it}^-(\xi) \quad \forall t \in T \quad (10m) \\
& \rho_t^{MIN} \leq \rho_t^+, \quad \rho_t^- \leq \rho_t^{MAX} \quad \forall t \in T \quad (10n) \\
& \Pi_i^{END} = \sum_{t \in T} (P_t^{DA} \cdot x_{it} + \mathbb{E} [P_t^{RT}(\xi) \cdot y_{it}^+(\xi) - P_t^{PN}(\xi) \cdot y_{it}^-(\xi) + \rho_t^+ \cdot d_{it}^+(\xi) - \rho_t^- \cdot d_{it}^-(\xi)]) \quad \forall i \in I \quad (10o) \\
& \Pi_i^{IND} \leq \Pi_i^{END} \quad \forall i \in I \quad (10p) \\
& \sum_{i \in I} \Pi_i^{END} \leq \Pi^{HOL} \quad (10q)
\end{aligned}$$

4.2 Price as DV - Market Clearing ver.

$$\begin{aligned}
\max \quad & \sum_{i \in I} \sum_{t \in T} (P_t^{DA} \cdot x_{it} + \mathbb{E} [P_t^{RT}(\xi) \cdot y_{it}^+(\xi) - P_t^{PN}(\xi) \cdot y_{it}^-(\xi) + \rho_t \cdot d_{it}^+(\xi) - \rho_t \cdot d_{it}^-(\xi)]) \quad (11a) \\
\text{s.t.} \quad & R_{it}(\xi) - x_{it} = y_{it}^+(\xi) - y_{it}^-(\xi) + d_{it}^+(\xi) - d_{it}^-(\xi) + z_{it}^C(\xi) - z_{it}^D(\xi) \quad \forall i \in I, t \in T \quad (11b) \\
& R_{it}(\xi) + z_{it}^D(\xi) \geq y_{it}^+(\xi) + d_{it}^+(\xi) + z_{it}^C(\xi) \quad \forall i \in I, t \in T \quad (11c) \\
& z_{i,t+1}(\xi) = z_{it}(\xi) + z_{it}^C(\xi) - z_{it}^D(\xi) \quad \forall i \in I, t \in T \quad (11d) \\
& z_{it}^D(\xi) \leq z_{it}(\xi), \quad z_{it}^C(\xi) \leq K_i - z_{it}(\xi), \quad 0 \leq z_{it}(\xi) \leq K_i \quad \forall i \in I, t \in T \quad (11e) \\
& y_{it}^+(\xi) \leq M\phi_{it}^1(\xi), \quad y_{it}^-(\xi) \leq M(1 - \phi_{it}^1(\xi)) \quad \forall i \in I, t \in T \quad (11f) \\
& y_{it}^-(\xi) \leq M\phi_{it}^2(\xi), \quad z_{it}^C(\xi) \leq M(1 - \phi_{it}^2(\xi)) \quad \forall i \in I, t \in T \quad (11g) \\
& z_{it}^C(\xi) \leq M\phi_{it}^3(\xi), \quad z_{it}^D(\xi) \leq M(1 - \phi_{it}^3(\xi)) \quad \forall i \in I, t \in T \quad (11h) \\
& d_{it}^+(\xi) \leq M\phi_{it}^4(\xi), \quad d_{it}^-(\xi) \leq M(1 - \phi_{it}^4(\xi)) \quad \forall i \in I, t \in T \quad (11i) \\
& z_{it}^C(\xi) \leq M\phi_{it}^5(\xi), \quad d_{it}^-(\xi) \leq M(1 - \phi_{it}^5(\xi)) \quad \forall i \in I, t \in T \quad (11j) \\
& y_{it}^+(\xi) \leq M\phi_{it}^6(\xi), \quad d_{it}^-(\xi) \leq M(1 - \phi_{it}^6(\xi)) \quad \forall i \in I, t \in T \quad (11k) \\
& y_{it}^-(\xi) \leq M\phi_{it}^7(\xi), \quad d_{it}^+(\xi) \leq M(1 - \phi_{it}^7(\xi)) \quad \forall i \in I, t \in T \quad (11l) \\
& \sum_{i \in I} d_{it}^+(\xi) = \sum_{i \in I} d_{it}^-(\xi) \quad \forall t \in T \quad (11m) \\
& \rho_t^{MIN} \leq \rho_t \leq \rho_t^{MAX} \quad \forall t \in T \quad (11n) \\
& \Pi_i^{END} = \sum_{t \in T} (P_t^{DA} \cdot x_{it} + \mathbb{E} [P_t^{RT}(\xi) \cdot y_{it}^+(\xi) - P_t^{PN}(\xi) \cdot y_{it}^-(\xi) + \rho_t^+ \cdot d_{it}^+(\xi) - \rho_t^- \cdot d_{it}^-(\xi)]) \quad \forall i \in I \quad (11o) \\
& \Pi_i^{IND} \leq \Pi_i^{END} \quad \forall i \in I \quad (11p)
\end{aligned}$$

5 Lagrangian Dual

5.1 Holistic Aggregation

$$\max \sum_{i \in I} \sum_{t \in T} \left(P_t^{DA} \cdot x_{it} + \frac{1}{S} \sum_{s \in S} (P_t^{RT}(\xi_s) \cdot y_{it}^+(\xi_s) - P_t^{PN}(\xi_s) \cdot y_{it}^-(\xi_s)) \right) \quad (12a)$$

$$\text{s.t. } R_{it}(\xi_s) - x_{it} = y_{it}^+(\xi_s) - y_{it}^-(\xi_s) + d_{it}^+(\xi_s) - d_{it}^-(\xi_s) + z_{it}^C(\xi_s) - z_{it}^D(\xi_s) \quad \forall i \in I, t \in T, s \in S \quad (12b)$$

$$R_{it}(\xi_s) + z_{it}^D(\xi_s) \geq y_{it}^+(\xi_s) + d_{it}^+(\xi_s) + z_{it}^C(\xi_s) \quad \forall i \in I, t \in T, s \in S \quad (12c)$$

$$z_{i,t+1}(\xi_s) = z_{it}(\xi_s) + \eta^C z_{it}^C(\xi_s) - \frac{1}{\eta^D} z_{it}^D(\xi_s) \quad \forall i \in I, t \in T, s \in S \quad (12d)$$

$$z_{it}^D(\xi_s) \leq z_{it}(\xi_s), \quad z_{it}^C(\xi_s) \leq K_i - z_{it}(\xi_s), \quad 0 \leq z_{it}(\xi_s) \leq K_i \quad \forall i \in I, t \in T, s \in S \quad (12e)$$

$$\sum_{i \in I} d_{it}^+(\xi_s) = \sum_{i \in I} d_{it}^-(\xi_s) \quad : \quad \lambda_t(\xi_s) \quad \forall t \in T, s \in S \quad (12f)$$

$$y_{it}^+(\xi_s) \leq M\phi_{it}^1(\xi_s), \quad y_{it}^-(\xi_s) \leq M(1 - \phi_{it}^1(\xi_s)) \quad \forall i \in I, t \in T, s \in S \quad (12g)$$

$$d_{it}^+(\xi_s) \leq M\phi_{it}^2(\xi_s), \quad d_{it}^-(\xi_s) \leq M(1 - \phi_{it}^2(\xi_s)) \quad \forall i \in I, t \in T, s \in S \quad (12h)$$

$$y_{it}^+(\xi_s) \leq M\phi_{it}^3(\xi_s), \quad d_{it}^-(\xi_s) \leq M(1 - \phi_{it}^3(\xi_s)) \quad \forall i \in I, t \in T, s \in S \quad (12i)$$

$$y_{it}^-(\xi_s) \leq M\phi_{it}^4(\xi_s), \quad d_{it}^+(\xi_s) \leq M(1 - \phi_{it}^4(\xi_s)) \quad \forall i \in I, t \in T, s \in S \quad (12j)$$

$$y_{it}^-(\xi_s) \leq M\phi_{it}^5(\xi_s), \quad z_{it}^C(\xi_s) \leq M(1 - \phi_{it}^5(\xi_s)) \quad \forall i \in I, t \in T, s \in S \quad (12k)$$

$$d_{it}^-(\xi_s) \leq M\phi_{it}^6(\xi_s), \quad z_{it}^C(\xi_s) \leq M(1 - \phi_{it}^6(\xi_s)) \quad \forall i \in I, t \in T, s \in S \quad (12l)$$

(??): Maximizing total expected revenue of the aggregation from day-ahead trading and real-time balancing.

(??): Energy balance equation ensuring the net energy flows regarding day-ahead commitment, real-time trading, internal trading, and battery operations equals the amount of generation.

(??): Feasibility constraint ensuring that total energy outflows do not exceed available energy.

(??): Battery state-of-charge dynamics with charging and discharge efficiency.

(??): Battery operational limits ensuring operations do not exceed currently available capacity.

(??): Internal trading balance constraint ensuring internal pool supply and demand are cleared, with dual variable $\lambda_t(\xi_s)$ representing the internal settlement price.

(??)-(??): Arbitrage control measures preventing the simultaneous buying and selling of energy within and among markets

(??)-(??): Arbitrage control measures prohibiting the purchase of energy for the purpose of charging the battery

5.2 Lagrangian Relaxed Holistic Aggregation

$$\max \sum_{i \in I} \sum_{t \in T} \left(P_t^{DA} \cdot x_{it} + \frac{1}{S} \sum_{s \in S} (P_t^{RT}(\xi_s) \cdot y_{it}^+(\xi_s) - P_t^{PN}(\xi_s) \cdot y_{it}^-(\xi_s)) \right) \quad (13a)$$

$$+ \sum_{s \in S} \sum_{t \in T} \lambda_t(\xi_s) \left(\sum_{i \in I} d_{it}^-(\xi_s) - \sum_{i \in I} d_{it}^+(\xi_s) \right)$$

$$\text{s.t. } R_{it}(\xi_s) - x_{it} = y_{it}^+(\xi_s) - y_{it}^-(\xi_s) + d_{it}^+(\xi_s) - d_{it}^-(\xi_s) + z_{it}^C(\xi_s) - z_{it}^D(\xi_s) \quad \forall i \in I, t \in T, s \in S \quad (13b)$$

$$R_{it}(\xi_s) + z_{it}^D(\xi_s) \geq y_{it}^+(\xi_s) + d_{it}^+(\xi_s) + z_{it}^C(\xi_s) \quad \forall i \in I, t \in T, s \in S \quad (13c)$$

$$z_{i,t+1}(\xi_s) = z_{it}(\xi_s) + \eta^C z_{it}^C(\xi_s) - \frac{1}{\eta^D} z_{it}^D(\xi_s) \quad \forall i \in I, t \in T, s \in S \quad (13d)$$

$$z_{it}^D(\xi_s) \leq z_{it}(\xi_s), \quad z_{it}^C(\xi_s) \leq K_i - z_{it}(\xi_s), \quad 0 \leq z_{it}(\xi_s) \leq K_i \quad \forall i \in I, t \in T, s \in S \quad (13e)$$

$$y_{it}^+(\xi_s) \leq M\phi_{it}^1(\xi_s), \quad y_{it}^-(\xi_s) \leq M(1 - \phi_{it}^1(\xi_s)) \quad \forall i \in I, t \in T, s \in S \quad (13f)$$

$$d_{it}^+(\xi_s) \leq M\phi_{it}^2(\xi_s), \quad d_{it}^-(\xi_s) \leq M(1 - \phi_{it}^2(\xi_s)) \quad \forall i \in I, t \in T, s \in S \quad (13g)$$

$$y_{it}^+(\xi_s) \leq M\phi_{it}^3(\xi_s), \quad d_{it}^-(\xi_s) \leq M(1 - \phi_{it}^3(\xi_s)) \quad \forall i \in I, t \in T, s \in S \quad (13h)$$

$$y_{it}^-(\xi_s) \leq M\phi_{it}^4(\xi_s), \quad d_{it}^+(\xi_s) \leq M(1 - \phi_{it}^4(\xi_s)) \quad \forall i \in I, t \in T, s \in S \quad (13i)$$

$$y_{it}^-(\xi_s) \leq M\phi_{it}^5(\xi_s), \quad z_{it}^C(\xi_s) \leq M(1 - \phi_{it}^5(\xi_s)) \quad \forall i \in I, t \in T, s \in S \quad (13j)$$

$$d_{it}^-(\xi_s) \leq M\phi_{it}^6(\xi_s), \quad z_{it}^C(\xi_s) \leq M(1 - \phi_{it}^6(\xi_s)) \quad \forall i \in I, t \in T, s \in S \quad (13k)$$

5.3 Individual Replay

for each i:

$$\max \sum_{t \in T} \left(P_t^{DA} x_t + \frac{1}{S} \sum_{s \in S} (P_t^{RT}(\xi_s) y_t^+(\xi_s) - P_t^{PN}(\xi_s) y_t^-(\xi_s)) \right) + \sum_{s \in S} \sum_{t \in T} \lambda_t(\xi_s) (d_t^-(\xi_s) - d_t^+(\xi_s)) \quad (14a)$$

$$\text{s.t. } R_t(\xi_s) - x_t = y_t^+(\xi_s) - y_t^-(\xi_s) + d_t^+(\xi_s) - d_t^-(\xi_s) + z_t^C(\xi_s) - z_t^D(\xi_s) \quad \forall t \in T, s \in S \quad (14b)$$

$$R_t(\xi_s) + z_t^D(\xi_s) \geq y_t^+(\xi_s) + d_t^+(\xi_s) + z_t^C(\xi_s) \quad \forall t \in T, s \in S \quad (14c)$$

$$z_{t+1}(\xi_s) = z_t(\xi_s) + \eta^C z_t^C(\xi_s) - \frac{1}{\eta^D} z_t^D(\xi_s) \quad \forall t \in T, s \in S \quad (14d)$$

$$z_t^D(\xi_s) \leq z_t(\xi_s), \quad z_t^C(\xi_s) \leq K - z_t(\xi_s), \quad 0 \leq z_t(\xi_s) \leq K \quad \forall t \in T, s \in S \quad (14e)$$

$$y_t^+(\xi_s) \leq M \phi_t^1(\xi_s), \quad y_t^-(\xi_s) \leq M(1 - \phi_t^1(\xi_s)) \quad \forall t \in T, s \in S \quad (14f)$$

$$d_t^+(\xi_s) \leq M \phi_t^2(\xi_s), \quad d_t^-(\xi_s) \leq M(1 - \phi_t^2(\xi_s)) \quad \forall t \in T, s \in S \quad (14g)$$

$$y_t^+(\xi_s) \leq M \phi_t^3(\xi_s), \quad d_t^-(\xi_s) \leq M(1 - \phi_t^3(\xi_s)) \quad \forall t \in T, s \in S \quad (14h)$$

$$y_t^-(\xi_s) \leq M \phi_t^4(\xi_s), \quad d_t^+(\xi_s) \leq M(1 - \phi_t^4(\xi_s)) \quad \forall t \in T, s \in S \quad (14i)$$

$$y_t^-(\xi_s) \leq M \phi_t^5(\xi_s), \quad z_t^C(\xi_s) \leq M(1 - \phi_t^5(\xi_s)) \quad \forall t \in T, s \in S \quad (14j)$$

$$d_t^-(\xi_s) \leq M \phi_t^6(\xi_s), \quad z_t^C(\xi_s) \leq M(1 - \phi_t^6(\xi_s)) \quad \forall t \in T, s \in S \quad (14k)$$

6 Linear Decision Rule

6.1 No Arbitrage Control

$$\max \quad \sum_{i \in I} \sum_{t \in T} \left(P_t^{DA} \cdot x_{it} + \frac{1}{S} \sum_{s \in S} (P_t^{RT}(\xi_s) \cdot y_{it}^+(\xi_s) - P_t^{PN}(\xi_s) \cdot y_{it}^-(\xi_s)) \right) \quad (15a)$$

$$\text{s.t.} \quad R_{it}(\xi_s) - x_{it} = y_{it}^+(\xi_s) - y_{it}^-(\xi_s) + d_{it}^+(\xi_s) - d_{it}^-(\xi_s) + z_{it}^C(\xi_s) - z_{it}^D(\xi_s) \quad \forall i \in I, t \in T, s \in S \quad (15b)$$

$$R_{it}(\xi_s) + z_{it}^D(\xi_s) \geq y_{it}^+(\xi_s) + d_{it}^+(\xi_s) + z_{it}^C(\xi_s) \quad \forall i \in I, t \in T, s \in S \quad (15c)$$

$$z_{i,t+1}(\xi_s) = z_{it}(\xi_s) + \eta^C z_{it}^C(\xi_s) - \frac{1}{\eta^D} z_{it}^D(\xi_s) \quad \forall i \in I, t \in T, s \in S \quad (15d)$$

$$z_{it}^D(\xi_s) \leq z_{it}(\xi_s), \quad z_{it}^C(\xi_s) \leq K_i - z_{it}(\xi_s), \quad 0 \leq z_{it}(\xi_s) \leq K_i \quad \forall i \in I, t \in T, s \in S \quad (15e)$$

$$\sum_{i \in I} d_{it}^+(\xi_s) = \sum_{i \in I} d_{it}^-(\xi_s) \quad : \quad \lambda_t(\xi_s) \quad \forall t \in T, s \in S \quad (15f)$$

$$y_{it}^+(\xi_s) = y_{it,0}^+ + y_{it,R}^+ \cdot R_{it}(\xi_s) + y_{it,RT}^+ \cdot P_t^{RT}(\xi_s) + y_{it,PN}^+ \cdot P_t^{PN}(\xi_s) \quad \forall i \in I, t \in T, s \in S \quad (15g)$$

$$y_{it}^-(\xi_s) = y_{it,0}^- + y_{it,R}^- \cdot R_{it}(\xi_s) + y_{it,RT}^- \cdot P_t^{RT}(\xi_s) + y_{it,PN}^- \cdot P_t^{PN}(\xi_s) \quad \forall i \in I, t \in T, s \in S \quad (15h)$$

$$d_{it}^+(\xi_s) = d_{it,0}^+ + d_{it,R}^+ \cdot R_{it}(\xi_s) + d_{it,RT}^+ \cdot P_t^{RT}(\xi_s) + d_{it,PN}^+ \cdot P_t^{PN}(\xi_s) \quad \forall i \in I, t \in T, s \in S \quad (15i)$$

$$d_{it}^-(\xi_s) = d_{it,0}^- + d_{it,R}^- \cdot R_{it}(\xi_s) + d_{it,RT}^- \cdot P_t^{RT}(\xi_s) + d_{it,PN}^- \cdot P_t^{PN}(\xi_s) \quad \forall i \in I, t \in T, s \in S \quad (15j)$$

$$z_{it}^C(\xi_s) = z_{it,0}^C + z_{it,R}^C \cdot R_{it}(\xi_s) + z_{it,RT}^C \cdot P_t^{RT}(\xi_s) + z_{it,PN}^C \cdot P_t^{PN}(\xi_s) \quad \forall i \in I, t \in T, s \in S \quad (15k)$$

$$z_{it}^D(\xi_s) = z_{it,0}^D + z_{it,R}^D \cdot R_{it}(\xi_s) + z_{it,RT}^D \cdot P_t^{RT}(\xi_s) + z_{it,PN}^D \cdot P_t^{PN}(\xi_s) \quad \forall i \in I, t \in T, s \in S \quad (15l)$$

6.2 dd